LEVELHULL · CONCEPT OPTIONS

LevelHull concept options

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DOCUMENT CONTROL

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REVISION HISTORY

Version	Date	Author	Change
0.2	2026-10-25	Amish Chadha	Option A cost USD 694 and STEM rows S32, S34 after Amish's decisions of 2026-10-25; D-O1 still open (no pick)
0.1	2026-10-24	Amish Chadha	First issue (kit 1.9.1 upgrade): the existing design as option A, external sponsons (B) and double outrigger floats (C), each modelled at D3; scored against LVH-URS-001

LevelHull concept options

Option A, the existing design (foam modules high inside the hull), scores 172 of 215; option B (external foam sponsons) scores 163, only 5.2 % behind, so **both go to Amish without a pick** (STANDARDS section 21). Option C (double outrigger floats) scores 148 and is not put forward. The repo stays on option A, its current design, until Amish decides: this is **Proposed, awaiting Amish** (LVH-DEC-001 D-O1, LVH-DDR-004). Method: STANDARDS section 21. Numbers: `docs/04-calcs/stem_check.py` (`options_results.csv`), LVH-CAL-001 and `cad/fidelity.json`.

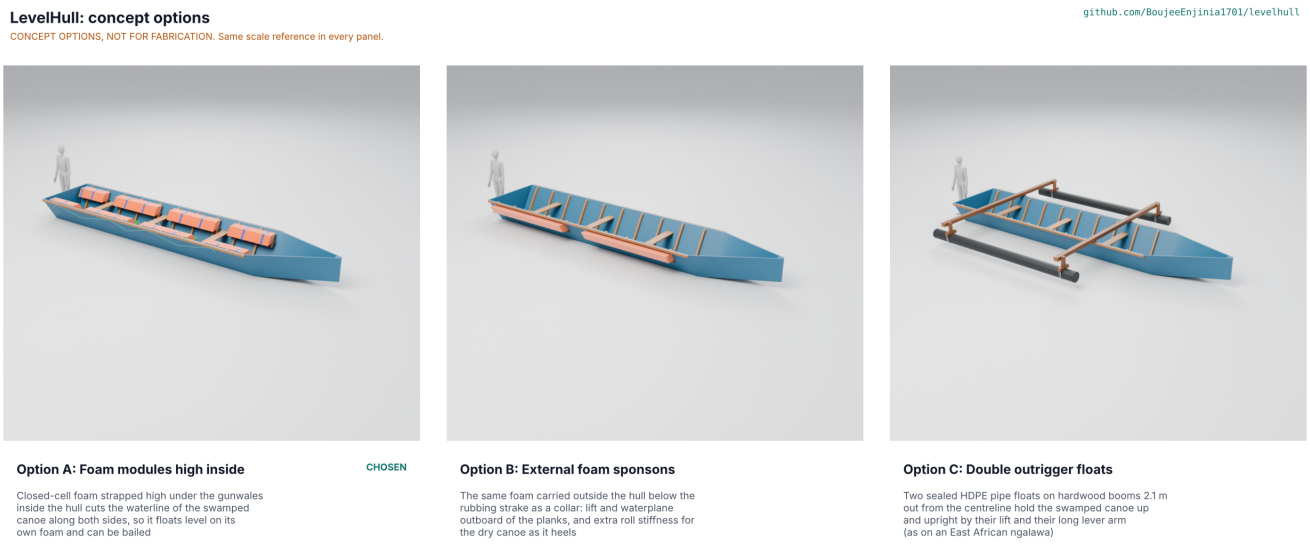


Figure 1. The three options, each modelled at detail level D3 in `cad/src/model.py`, drawn at one scale with a 1.75 m person (rendered with `python .kit/options.py` in Blender Cycles).

> **Safety:** Every option is flotation for a boat, not for people, and none is a substitute for life jackets. Each would be swamp-tested in sheltered, shallow water with a safety boat and everyone in a life jacket before use.

The options

Option A: foam modules high inside (the design)

Closed-cell foam strapped high under the gunwales inside the hull cuts the waterline of the swamped canoe along both sides, so it floats level on its own foam and can be bailed. Eight modules of four 50 mm polyethylene foam strips in sewn tarpaulin covers, 220 x 200 mm, 4.78 m a side, sit on hardwood shelf battens coach-screwed to the frames and are held down by webbing straps with D-ring buckles; grab lines run along both sides through eye bolts in the top plank. This is the constructable design of LVH-DDR-002 and LVH-DDR-003, unchanged except for the chock screw fix of LVH-DEC-001 D-S1. Swamped at the design load: 92 mm lowest freeboard, 761 kg m per radian, 2.45 times the design load in reserve, still afloat with the largest module lost. It takes 8.4 % of the inside volume, high along the sides. Key risks: cover and strap life in the sun (S10), owners accepting the lost space. Needs from a partner: hull and load surveys, a swamp test.

Option B: external foam sponsons

The same foam carried outside the hull below the rubbing strake as a collar: lift and waterplane outboard of the planks, and extra roll stiffness for the dry canoe as it heels. Two sponsons a side, each 2.39 m long, of the same 220 x 200 mm foam stack in a tarpaulin sleeve whose doubled flanges are clamped to the planking by an outside hardwood shelf batten (below) and a hardwood clamp strip (above, up to the rubbing strake), both through-bolted with M8 bolts through the planks into every frame. Swamped: 158 mm lowest freeboard, 1,546 kg m per radian, 1.9 deg with all four crew on one side; nothing inside the hull. Against it: 32 new bolt holes through the planks (150 mm above the loaded waterline), a beam of 1.93 m over the sponsons, covers exposed to the beach, to rafting alongside and to nets hauled over the side (S27), and, as drawn, losing one sponson (half a side) puts the gunwale 32 mm under. Needs from a partner: a trial of abrasion on the landing beach and of net handling over the sponsons.

Option C: double outrigger floats

Two sealed HDPE pipe floats on hardwood booms 2.1 m out from the centreline hold the swamped canoe up and upright by their lift and their long lever arm, as on an East African ngalawa. Two 120 x 90 mm hardwood booms cross the gunwales over the aft and forward thwarts on saddle blocks, bolted down through the rubbing strakes; a strut with a saddle foot at each boom end carries a 4.46 m float of 250 mm PE100 SDR 26 pipe with butt-fused end caps, held by stainless U-bolts. Swamped: 160 mm lowest freeboard and very high roll stiffness (6,300 kg m per radian); the dry canoe also resists capsize, which A and B do not. Against it: a 4.45 m beam at crowded landing sites, booms and floats in the way of nets hauled over the side, 129 kg of kit, butt fusion of the caps needs a fusion machine, and with one float lost the canoe capsizes. Needs from a partner: a fishery that already sails outrigger canoes (coastal, not the planked lake canoes of the reference).

First-order numbers

Table 1. Options compared (reference canoe, fresh water, design net load 118 kg; option A from LVH-CAL-001, B and C from `stem_check.py` with the same method; part counts and model masses from `cad/fidelity.json`).

Quantity	Option A	Option B	Option C
Estimated parts cost per canoe (USD)	694 (priced BOM, four bailers)	about 620 (estimate from the BOM's own unit prices; bolts priced as BOM line 3)	about 220 known (timber, bolts, bailers) plus 8.9 m of 250 mm HDPE pipe, price to assess
Kit mass (kg)	68.9 (CAL); 98.8 in the model	about 70 (estimate); 98.7 in the model	128.8 (estimate); 159.9 in the model
Buoyancy volume (m3)	0.42	0.42	0.44
Swamped lowest freeboard (mm)	92	158	160
Swamped roll stiffness (kg m per radian)	761	1,546	6,313
Four crew on one side: heel, low-side freeboard	3.9 deg, 44 mm	1.9 deg, 123 mm	0.5 deg, 132 mm
Largest buoyancy unit lost	Floats, 25 mm	Gunwale 32 mm under	Capsizes
Inside volume taken	8.4 %	None	None (booms cross above the gunwale)
Overall beam	1.68 m (unchanged)	1.93 m	4.45 m
New holes in the hull	8, in the top plank	32 through the planks, 150 mm above the loaded waterline	16, through the rubbing strakes
Parts in the model (D3)	209	124	124
STEM check	Within known science; 4 field items open; brim-full freeboard and bailing (S32, S34) closed by Amish's decisions D-O2 and D-O3 of 2026-10-25	Within; abrasion open; holes below the swamped waterline	Within

All three options are sealed hulls, so the brim-full condition just after swamping (LVH-STM-001 S2, S32 to S34) applies to each of them alike and does not change the ranking. The model masses are higher than the calculated kit masses because the kit has no closed-cell foam material, so the foam is modelled at 120 kg/m3 (four times the real 30 kg/m3), and hardwood at 800 kg/m3. The cost of option B is an estimate built from the unit prices already in `bom/bom.csv`; option C's pipe price is not known and is left to assess, as no price is invented.

Scoring

Table 2. Weighted scoring (weights: High UR 3, Medium 2, Low 1; scores 1 to 5). UR-010 (cost) is scored in the cost row. UR-015 (life jacket message) and UR-016 (sizing method) score the same for every option and are left out.

Criterion	Source	Weight	A	B	C	Reason
Stays afloat with crew, motor, gear and catch	UR-001	3	4	5	5	Reserve 2.45 for A; B and C carry more at 50 mm
Gunwale clear of the water when swamped	UR-002	3	4	5	5	92, 158 and 160 mm
Floats upright and level	UR-003	3	4	5	5	Roll stiffness 761, 1,546 and 6,313 kg m per radian
Two crew can bail it out	UR-004	3	4	4	4	Same water to bail in each
Room to work, stow catch and handle nets	UR-005	3	3	5	2	A takes 8.4 % high on the sides; B nothing inside; C's booms cross the canoe and its floats sit where nets come in
Keeps its lift in water, sun and fuel	UR-006	3	4	3	4	A's covers are inside; B's are outside on the beach; C's PE100 pipe is tough
Stays fixed when full of water or rolled	UR-007	3	4	3	3	A's fixings at 5 times their share; B hangs on tarpaulin flanges; C's booms take wave loads (8.5 MPa against 16)
Fitted in a day with hand tools, no holes below the waterline	UR-008	3	5	3	2	A: 8 holes in the top plank; B: 32 holes in the planks; C: butt fusion needs a machine
Lasts a year of normal use	UR-009	2	3	2	4	Tarpaulin life decides A and B
Reserve for an extra person or a wet net	UR-011	2	4	5	5	B and C have more freeboard
Still floats with one unit damaged	UR-012	2	4	1	1	A: 25 mm; B: gunwale under; C: capsizes
Level with all crew on one side	UR-013	3	4	5	5	3.9, 1.9 and 0.5 deg
Out of the way when beaching, landing and fishing	UR-014	2	5	2	1	A inside the hull lines; B 1.93 m beam; C 4.45 m
Cost against the value-engineering target	UR-010	2	3	3	2	A USD 694, B about USD 620, C pipe unpriced and the most material
Buildable with local skills and tools		2	5	4	2	A and B: carpentry and sewing; C needs HDPE fusion
Safety		3	4	4	3	C also stops the dry canoe capsizing, but one lost float capsizes it and the booms are a hazard alongside
Maintenance		1	4	3	4	B's external covers wear first
Weighted total (of 215)			**172**	**163**	**148**	A 80 %, B 76 %, C 69 %

Recommendation

A and B are within 10 % of each other (163 against 172, 5.2 %), so both go to Amish without a pick, as the standard asks. The case for each:

- **Keep option A (the design).** It is already constructable, priced and planned; it keeps everything inside the hull lines, needs only eight holes in the top plank and survives the loss of its largest module. Its weakness is the space it takes and the lowest freeboard of the three (92 mm, still above the 50 mm target).
- **Change to option B.** The same foam outside the planks gives two-thirds more freeboard and twice the roll stiffness and takes no space inside. It needs 32 bolt holes through the planks, widens the canoe to 1.93 m, exposes the covers to beach and nets, and as drawn fails R12 if one sponson is lost (shorter sponsons, three or four a side, would cure that at more cost).

Best idea to graft from the runner-up. B's numbers show that the same foam gives more freeboard when it sits lower, as long as it still cuts the waterline: in B the foam starts 360 mm up, in A 420 mm. Lowering A's modules and battens by about 40 mm would raise its swamped freeboard and keep everything inside, at some cost in clear width and roll stiffness. That is a change to a design dimension and goes to Amish with D-O1; it is not made here. From C, nothing is grafted: its intact stability needs the outriggers themselves.

The choice is recorded as "Proposed, awaiting Amish" in LVH-DEC-001 (D-O1) and argued in LVH-DDR-004. Until he decides, the drawing, priced BOM and build plan describe option A.